

PRACTICE TEST

16



16

TOUGHEST QUESTIONS #2

READING AND WRITING

DIRECTIONS

This section features some of the most challenging questions from the official 2025 SAT exams. Each item is designed to test advanced reading and writing skills. You will encounter one or more passages per question, which may be accompanied by a table or graph. Read the passage(s) and question carefully, and then select the best answer. All questions are multiple-choice with four options, and only one answer is correct.

Please take as much time as you find necessary to complete the test thoroughly.

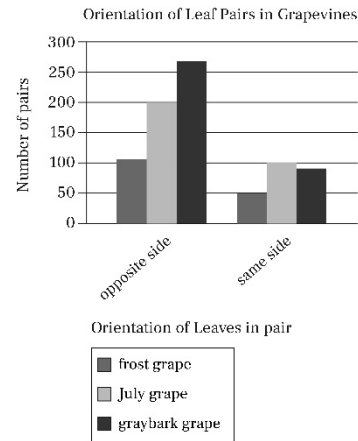
1

Walt Whitman's *Leaves of Grass* first appeared in 1855 as a slim collection of twelve poems, but Whitman would **revise and expand it** **substantially** over the next four decades. **These extensive** _____ the addition of hundreds of new poems, the removal of some existing ones, and the insertion of prefatory material, **reflected** the poet's evolving literary perspective and experience of the US Civil War.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) changes would include
- B) changes, include
- C) **changes included**
- D) changes, including

2



Auxins are a class of hormones that influence plant growth, including leaf orientation (the tendency of leaves to be larger on one side of their long central axis than the other). University of California, Berkeley biologist Ciera Martinez and colleagues noted that in certain plants in which leaves grow in pairs, auxins will typically be concentrated in opposite sides of each leaf in the pair (e.g., on the left side of one leaf in the pair and the right side of the other). Accordingly, they hypothesized that **paired leaves** should tend to **show opposite-side orientation**, and tested their hypothesis by **examining paired leaves from several species of grapevines**.

Which choice best describes data from the graph that **support Martinez and colleagues' hypothesis**?

- A) The number of leaf pairs showing opposite-side orientation is fairly high in the July grape, but not as high as in the graybark grape.
- B) Although the number of leaf pairs showing same-side orientation is fairly high in the July grape, it is much lower in both the frost grape and graybark grape.
- C) Although the exact ratio varies by species, the graybark grape, frost grape, and July grape **all show more leaf pairs with opposite-side orientations** than with same-side orientations.
- D) In the graybark grape, frost grape, and July grape, all the leaf pairs show opposite-side orientation.

3

Researchers have noted that people with clinically typical hearing perceive sounds they believe to be meaningful as quieter than sounds of the same volume that they believe to be noise (i.e., meaningless to the listener). In a recent study, Antonia Olivia Dolan and colleagues allowed participants to listen to (and adjust the volume of) recordings of music in popular genres like acoustic folk and orchestral, as well as recordings of nature sounds. The researchers noted that participants may have treated the nature sounds as noise, which suggests that if a participant was exposed to José González's "Heartbeats" and the nature sounds at a volume of 61.5 decibels, the participant likely would have _____.

Which choice most logically completes the text?

- A) adjusted the volume of the nature sounds to be greater than 61.5 decibels.
- B) believed that neither the nature sounds nor "Heartbeats" were at a volume of 61.5 decibels.
- C) experienced the nature sounds as louder than "Heartbeats" even though they were not.
- D) perceived the nature sounds and "Heartbeats" to be comparably meaningful despite perceiving the music to be quieter.

4

The jade hawkmoth, a large-bodied moth, defends itself against Brandt's myotis and other insect-eating bats, which use echolocation to hunt, by emitting ultrasonic clicks that can, for instance, signal the moth's unpleasant taste. To investigate moths' defensive ultrasound—which researchers had thought was exclusive to tiger moths, hawkmoths, and one species of geometer moths—Jesse R. Barber et al. recorded the responses of moths from 252 genera, representing most families of large-bodied moths, to audio playback of bat echolocation. The researchers found that 52 of the genera, including several genera belonging to the geometer family, produced defensive ultrasonic clicks. This result suggests that _____.

Which choice most logically completes the text?

- A) unlike the 52 moth genera that emit ultrasonic clicks, most moth genera have likely not developed defenses specifically against bat attacks.
- B) some genera of large-bodied moths may use ultrasonic signaling for purposes other than avoiding capture by predators such as Brandt's myotis.
- C) ultrasound production is only one of a diverse range of effective strategies moths employ to evade bat attacks.
- D) anti-bat ultrasound production may be a more prevalent defense strategy among large-bodied moths than previously known to researchers.

5

Scholars cite *Men of Maize*, the 1949 novel by Guatemalan author Miguel Ángel Asturias, as a foundational text of magical realism, the Latin American style of fiction in which antirealistic plot devices—often borrowed from the folkloric traditions of Indigenous and colonial societies in the Americas—are deployed in an otherwise realistic mode of representation typical of the modern novel. This style has exerted a decisive influence on authors around the world, including Orhan Pamuk, whose 2001 novel *My Name Is Red* resembles classic magical realist novels in its juxtaposition of literary realism with folklore—namely, that of Turkey.

Which quotation from a literary scholar would most directly support the claim in the underlined portion of the text?

- A) “Like many works in the Latin American magical realist tradition, *My Name Is Red* is indebted to antirealistic elements in the folkloric tradition of Turkey.”
- B) “Although Turkish folklore clearly informs the style and occasionally antirealistic plot of *My Name Is Red*, the novel also shows the inarguable influence of the magical realist tradition of Latin America.”
- C) “While *My Name Is Red* alternates between realistic and antirealistic modes of representation, details suggesting the influence of Turkish folklore nevertheless occur throughout the novel.”
- D) “The logic of the realistic plot of *My Name Is Red* is repeatedly and productively disrupted by the presence of imagery and situations drawn from Turkish folklore.”

6

The parks of San Jose, California, seem to be making people happier. In a 2022 study, researchers _____ for a relationship between the physical location in which a social media post was created and the content of that post analyzed geotagged social media posts from San Jose. The team found that posts from the city’s parks contained more words associated with happiness than other posts did.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) were looking
- B) looked
- C) looking
- D) had looked

7

For Canada, a member of the North Atlantic Treaty Organization (NATO) since 1949, NATO’s principle of collective defense confers both benefits and _____ organization’s many members, nations as disparate as the US and Slovenia, are all bound to defend Canada, the reverse is also true.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) obligations: while the
- B) obligations; the
- C) obligations, while the
- D) obligations. The

8

Jordan's constitution, enacted in 1952, contains just one of the six constitutional features that enhance judicial independence, as identified by legal scholars James Melton and Tom Ginsburg. After 1985, such de jure judicial independence became more prevalent, as exemplified in Croatia's constitution, enacted in 1991 and _____ four of these features.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) contained
- B) containing
- C) had contained
- D) contains

9

The following text is from Georgia Douglas Johnson's 1918 poem "Peace."

I rest me deep within the wood,
Drawn by its silent call,
 Far from the throbbing crowd of men,
 On nature's breast I fall.

As used in the text, what does the word "drawn" most nearly mean?

- A) Unnoticed
- B) Depicted
- C) Enticed
- D) Envisioned

10

The following text is from José Rizal's 1891 novel *The Reign of Greed* (translated by Charles Derbyshire in 1912).

A lady accompanied by her husband entered at that moment and took her place in one of the two vacant boxes. She had the air of a queen and gazed disdainfully at the whole house, as if to say, "I've come later than all of you, you crowd of upstarts and provincials. I've come later than you!" There are persons who go to the theater like the contestants in a mule-race: the last one in, wins, and we know very sensible men who would ascend the scaffold rather than enter a theater before the first act.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It conveys the urgency some theater patrons feel to be the first ones to arrive at a performance.
- B) It emphasizes a character's desire to be an actor.
- C) It helps explain why some people dislike attending theatrical performances.
- D) It illustrates the lady's contemptuous attitude toward the other people in the theater.

11

While researching a topic, a student has taken the following notes:

- Freezers and other cooling technologies use chemical compounds called refrigerants to absorb heat and release cold air.
- When a refrigerant absorbs heat, it undergoes vaporization, changing from a liquid to a gas.
- When it cools, it undergoes condensation, changing from a gas to a liquid.
- The refrigerant fluoromethane is a hydrofluorocarbon (HFC).
- HFCs are composed of the elements hydrogen, fluorine, and carbon.

The student wants to describe the role of refrigerants in freezers. Which choice effectively uses relevant information from the notes to accomplish this goal?

- A) Freezers can absorb heat and release cold air through vaporization and condensation.
- B) The refrigerant fluoromethane, used in freezers, is made up of hydrogen, fluorine, and carbon.
- C) Used in freezers, refrigerants such as fluoromethane can change from a gas to a liquid and back.
- D) Freezers use refrigerants like fluoromethane to absorb heat and release cold air.

12

The star Wezen is unvaryingly bright, consistently ranking as one of the brightest stars in the sky (Wezen ranked 36th on a recent list). By contrast, the brightness of the red giant Betelgeuse varies considerably. Indeed, between April and June 2023, Betelgeuse became an astonishing 40% brighter; _____ its ranking rose from 10th to 7th.

as a result

Which choice completes the text with the most logical transition?

- A) in sum,
- B) in turn,
- C) specifically,
- D) for example,

13

Helical _____ widely understood to confer stability and efficiency in the locomotion of a variety of microscopic organisms—including bacteria, eukaryotic algae, and ciliates—bestows similar advantages, albeit via different propulsive modes, to larger oceanic macroplankton, such as salps.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) swimming is
- B) swimming, is
- C) swimming has been
- D) swimming,

14

As disparate as their works may seem, commonalities exist between Antonello da Messina, Italian Renaissance painter of the Venetian _____ Flemish mannerist painter of the Antwerp School; and Jose Guerrero, American abstract painter of the New York School.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) School, Adriaen Brouwer,
- B) School; Adriaen Brouwer,
- C) School, Adriaen Brouwer;
- D) School Adriaen Brouwer,

15

With one in Finland and another in Australia, the observatory sites that form the Super Dual Auroral Radar Network—or SuperDARN, as space physicists like Mark Lester call it—number nearly five _____ across the globe, **the radars track ionospheric irregularities** and other geospace phenomena.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) **dozen. Located**
- B) dozen, located
- C) dozen, which are located
- D) dozen located

16

Bear River High School and Bryce Valley High School are two of several Utah _____ **enormous geoglyph** of the letters *BR* **overlooks** Bear River High, while a geoglyph of the letters *BV* overlooks Bryce Valley High.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) schools, that have their own hillside geoglyphs, and an
- B) schools that have their own hillside geoglyphs. An**
- C) schools, that have their own hillside geoglyphs and an
- D) schools that have their own hillside geoglyphs, an

17

The discoverers of the minor planet 713 Luscinia named it after the animal genus that includes nightingales. Most of the recently discovered minor planets, however, are given only an identification number, largely due to there being over 500,000 such bodies known at present, which makes the already challenging task of finding a unique name for each nearly _____.

Which choice completes the text with the most logical and precise word or phrase?

- A) substantial
- B) **insurmountable**
- C) unnecessary
- D) feasible

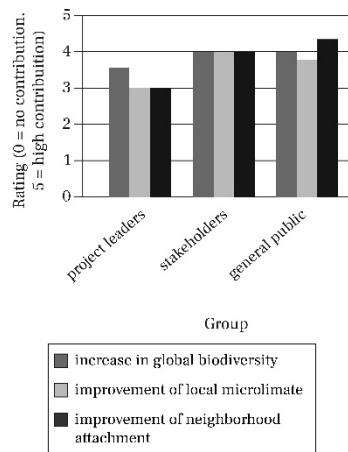
18

In the 1960s, Greece, Malta, and dozens of other countries were allocated unique country dialing codes to route international calls. **Most of the original two- or three-digit codes** _____ by the International Telegraph and Telephone Consultative Committee **later changed in response** to geopolitical circumstances or telecommunications needs.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) were assigned
- B) had been assigned
- C) **assigned**
- D) would be assigned

19



Urban agriculture is the practice of growing plant- or animal-based products in urban settings such as community gardens and rooftop farms. Esther Sanye-Mengual, Kathrin Specht, and their team surveyed three groups of people in Bologna, Italy—leaders of urban agriculture projects, stakeholders in urban agriculture (e.g., food researchers and urban farming associations), and the general public—to compare their views about the extent to which urban agriculture contributes to 25 social or ecological services that the team identified. The survey results show that, on average, project leaders rated urban agriculture as contributing less to _____.

Which choice most effectively uses data from the graph to complete the assertion?

- A) improving the local microclimate than to increasing global biodiversity.
- B) increasing global biodiversity than to providing tourist attractions.
- C) improving community members' sense of attachment to the neighborhood than to offering educational opportunities.
- D) increasing global biodiversity than to improving the local microclimate.

20

In their 2019 study of butterfly diversity at Rupa Lake in Nepal, researcher Bijaya Neupane and his colleagues counted one bluetail jester _____ to the Nymphalidae family; the bluetail jester (*Symbrenthia niphanda*) is known to be very rare.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) butterfly. Belonging
- B) butterfly belonging
- C) butterfly, belonging
- D) butterfly and belonging

21

Driven to sell as many paintings as possible, Alfred Hair, an influential figure among the landscape artists known as the Florida Highwaymen, pioneered "fast painting," a technique (which in part involved swift applications of paint) that many Highwaymen, including Isaac Knight, adopted. To conclude that this approach accounts for the ethereal qualities now synonymous with the Highwaymen aesthetic is tempting but inaccurate, as Hair's methods weren't universally practiced by his affiliates: Roy McLendon, for example, painted with greater deliberateness but achieved the same effects.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It details evidence that contradicts a claim mentioned earlier in the text about a long-standing disagreement within a group of artists.
- B) It explains how an artist mentioned earlier in the text developed a distinctive style by adapting a particular approach to painting originated by his colleagues.
- C) It considers and rebuts an interpretation of the effect of a painting technique mentioned earlier in the text on the perception of work by a group of artists.
- D) It establishes contrast between the aesthetic qualities of works by artists who were central to a movement introduced earlier in the text and those of an artist who was more peripheral to that movement.

22

The *Clouds* is a 423 BCE play by Aristophanes, originally written in ancient Greek. At the time, professional intellectuals called sophists taught customers rhetorical techniques to use in public speaking, along with providing instruction in other subjects. In the play, Aristophanes satirizes sophists as teaching students how to be persuasive without regard for what is morally good, as seen when the character _____.

Which choice most effectively uses a quotation from a translation of the play to illustrate the claim?

- A) Socrates, a sophist, says that he studies astronomy while seated in a basket hanging a few feet off the ground because “I should not have rightly discovered things celestial if I had not suspended the intellect, and mixed the thought in a subtle form with kindred air.”
- B) Strepsiades encourages his son to learn to be a sophist, saying, “Go, I entreat you, dearest of men, go and be taught.”
- C) Socrates, a sophist, says to a new customer, “Come now; what do you wish to learn first of those things in none of which you have ever been instructed?”
- D) Strepsiades hires a sophist to teach his son and says to the sophist, “But see that he learns those two [styles of argument]; the better, whatever it may be; and the worse, which, by maintaining what is unjust, overturns the better. If not both, at any rate the unjust one by all means.”

23

The 1960 founding of the Mexican American Political Association and the 1946 *Mendez v. Westminster* court decision are regarded as important events in US civil rights _____ former establishing a Latino rights advocacy group and the latter legally affirming the rights of Latino students.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) history the
- B) history, the
- C) history, as the
- D) history. The

MATHEMATICS

DIRECTIONS

This section features some of the most challenging math questions from the official 2025 SAT exams. The questions assess a range of advanced mathematical skills.

Please take as much time as you find necessary to complete the test thoroughly.

1

In the xy -plane, a circle has center C with coordinates (h, k) . Points A and B lie on the circle. Point A has coordinates $(h + 1, k + \sqrt{230})$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

- A) $231\sqrt{3}$
- B) $231\sqrt{2}$
- C) $2\sqrt{230}$
- D) $\sqrt{462}$

2

In triangle ABC and triangle DEF , sides AB and DE each have a side length of 10 inches, and angles A and D each have an angle measure of 40° . Which of the following additional pieces of information is (are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

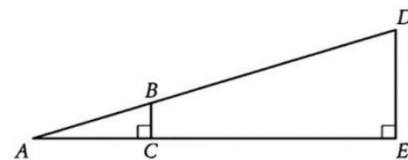
- I. The measures of angles B and C are equal.
 - II. The lengths of sides AC and DF are equal.
 - III. The lengths of sides BC and EF are equal.
- A) I is sufficient, but II and III are not.
 - B) II is sufficient, but I and III are not.
 - C) III is sufficient, but I and II are not.
 - D) II is sufficient and III is sufficient, but I is not.

3

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

- A) $-14x + 1,540$
- B) $14x - 320$
- C) $-28x + 1,720$
- D) 360

4



Note: Figure not drawn to scale.

In the figure shown, $AB = \sqrt{89}$ units, $AC = 8$ units, and $CE = 24$ units. What is the area, in square units, of triangle ADE ?

5

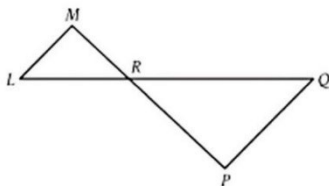
The volume of a right rectangular prism with a square base is 539 cubic centimeters. If the area of each of the four lateral faces is 77 square centimeters, what is the height, in centimeters, of the prism?

- A) 77
- B) 49
- C) 11
- D) 7

6

A polygon has exactly 87 sides. If the measure of each of the 87 interior angles of this polygon is $(180p)^\circ$, what is the value of p ?

7



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

- A) $\frac{579}{46}$
- B) $\frac{144}{7}$
- C) 63
- D) $\frac{441}{4}$

8

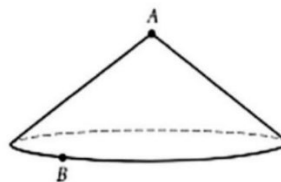
Right rectangular prism X is similar to right rectangular prism Y . The surface area of right rectangular prism X is 52 square centimeters (cm^2), and the surface area of right rectangular prism Y is $3,328cm^2$. The volume of right rectangular prism X is 10.5 cubic centimeters (cm^3). What is the volume, in cm^3 , of right rectangular prism Y ?

9

A circle is inscribed in a square such that the circumference of the circle touches the midpoint of each side of the square. The length of the diagonal of the square is 228 units. What is the area, in square units, of the circle?

- A) $6,498\pi$
- B) $12,996\pi$
- C) $25,992\pi$
- D) $51,984\pi$

10



Note: Figure not drawn to scale.

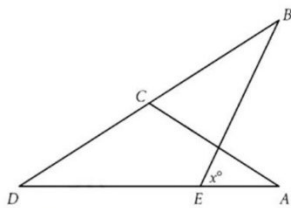
For the right circular cone shown, B is a point on the circumference of the base, and the length of segment AB (not shown) is 84 centimeters. If the height of the cone is 42 centimeters and the volume of the cone is $k\pi$ cubic centimeters, what is the value of k ?

11

In triangle FGH and triangle KLM , the measures of angles G and L are each 60° . The lengths of $\overline{GH}$ and $\overline{LM}$ are each 34 centimeters, and $\frac{GH}{GF} = \frac{LM}{LK}$. Which additional piece of information would be necessary to prove that triangle FGH is congruent to triangle KLM ?

- A) The lengths of $\overline{FG}$ and $\overline{KM}$ are each 17 centimeters.
- B) The measures of angles H and M are each 30° .
- C) The measures of angles F and K are each 90° .
- D) No additional information is necessary.

12



Note: Figure not drawn to scale.

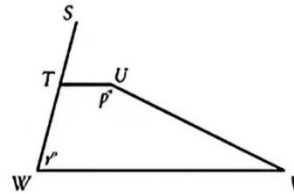
In the figure, $AC = CD$. The measure of angle EBC is 29° , and the measure of angle ACD is 108° . What is the value of x ?

13

A cube has an edge length of 82 inches. A solid sphere with a radius of 41 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

- A) 262,672
- B) 288,532
- C) 334,846
- D) 546,087

14



Note: Figure not drawn to scale.

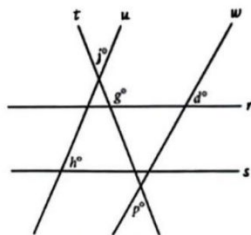
In the figure shown, point T lies on segment SW , and segment TU is parallel to segment VW . The measure of angle V is 26° , and the measure of angle STU is 74° . What is the value of $p - r$?

15

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $(5, 2)$, $(5, 4)$, and $(7, 2)$, respectively. Triangle LMN has vertices L , M , and N at $(5, 2)$, $(5, 4 + k)$, and $(7 + k, 2)$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

- A) $(90 - (t - k))^\circ$
- B) $(90 - (t + k))^\circ$
- C) $(90 - t)^\circ$
- D) $(90 + k)^\circ$

16

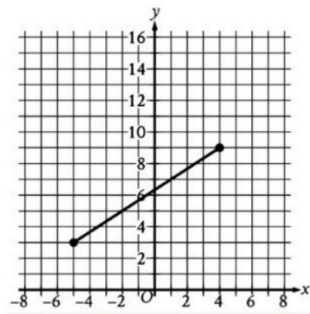


Note: Figure not drawn to scale.

In the figure, parallel lines r and s are intersected by lines t , u , and w . If $d = 42$, $g = 143$, and $h > d$, which statement about j and p must be true?

- A) $j = p$
- B) $j + p = 90$
- C) $j > p$
- D) $j < p$

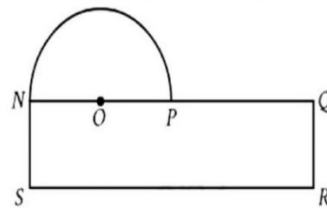
17



The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $48\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

- A) 16
- B) 32
- C) $3\sqrt{13}$
- D) $6\sqrt{13}$

18



Note: Figure not drawn to scale.

In the given figure, $\overline{NP}$ is the diameter of semicircle O . Angles Q , R , and S are right angles, and point P is the midpoint of $\overline{NQ}$. If $PQ = 52$ feet and $NS = 18$ feet, the area of the figure can be expressed as $a\pi + b$ square feet, where a and b are integers. What is the value of $a - b$?

19

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the bases of each of the two pyramids is $100cm^2$. Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2 ?

- A) 209
- B) 220
- C) 367
- D) 551

20

For two acute angles, $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 27$ and $4x + 3$, respectively. What is the value of x ?

- A) 8
- B) 12
- C) 30
- D) 63

21

A circle in the plane has its center at $(3, 7)$. Line t is tangent to this circle at the point $(a, -4)$, where a is a constant. The slope of line t is $\frac{5}{4}$. What is the value of a ?

- A) $-\frac{43}{4}$
- B) $-\frac{29}{5}$
- C) $\frac{59}{5}$
- D) $\frac{67}{4}$

22

	Volume (cubic units)
Right circular cylinder A	72π
Right circular cylinder B	$4,608\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 3 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$?

(The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

AVISO IMPORTANTE:

Caso você identifique questões com erros durante o simulado, por favor, envie um e-mail para a nossa equipe operacional, através do e-mail operacional@applyprep.com. As correções serão feitas e disponibilizadas em até 48 horas após o recebimento do seu e-mail.

Agradecemos pela sua compreensão e colaboração!

Boa prova :)